59. You are given normal boiling points and standard enthalpies of vapourisation. Calculate the entropy of vapourisation of liquids listed below.

S. No	Liquid	Boiling points (^{0}C)	ΔH (kJ mol^{-1})
1.	Ethanol	78.4	+ 42.4
2.	Toluene	110.6	+ 35.2

60. For the reaction $Ag_2O(s) \rightarrow 2Ag(s) + \frac{1}{2} O_2(g)$: ΔH = 30.56 kJ mol^{-1} and ΔS=6.66JK^{-1}mol^{-1}(at 1 atm). Calculate the temperature at which ΔG is equal to zero. Also predict the direction of the reaction (i) at this temperature and (ii) below this temperature.

61. What is the equilibrium constant Keq for the following reaction at 400K.

$2NOCl\ (g) \rightleftharpoons 2NO(g) + Cl_2(g)$,

given that $\Delta H^0 = 77.2$kJ mol^{-1} ;

and ΔS^0=122JK^{-1}mol^{-1}.

62. Cyanamide (NH_2CN) is completely burnt in excess oxygen in a bomb calorimeter, ΔU was found to be −742.4 kJ mol−1, calculate the enthalpy change of the reaction at 298K. $NH_2CN(s) + \frac{3}{2} O_2(g) \rightarrow N_2(g) + CO_2(g) + H_2O(l)$ ΔH= ?

63. Calculate the enthalpy of hydrogenation of ethylene from the following data. Bond energies of C – H, C – C , C = C and H – H are 414, 347, 618 and 435 kJ mol^{-1}.

64. Calculate the lattice energy of $CaCl_2$ from the given data

$Ca(s) + Cl_2(g) \rightarrow CaCl_2(s)$ ΔH_f^0 = – 795 kJ mol^{-1}

Sublimation : $Ca(s) \rightarrow Ca(g)$ ΔH_1^0 = + 121 kJ mol^{-1}

Ionisation : $Ca(g) \rightarrow Ca^{2+}(g) + 2e^-$ ΔH_2^0 = + 2422 kJ mol^{-1}

Dissociation : $Cl_2(g) \rightarrow 2Cl(g)$ ΔH_3^0 = + 242.8 kJ mol^{-1}

Electron affinity : $Cl(g) + e^- \rightarrow Cl^-(g)$ ΔH_4^0 = –355 kJ mol^{-1}

65. Calculate the enthalpy change for the reaction $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ from the following data.

$2Fe + \frac{3}{2} O_2 \rightarrow Fe_2O_3$; ΔH = –741 kJ

$C + \frac{1}{2} O_2 \rightarrow CO$; ΔH = –137 kJ

$C + O_2 \rightarrow CO_2$; ΔH = – 394.5 kJ

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